EXPERIMENT 5

Lamport Mutual Exclusion Algorithm:

lamport.py Python

```python
import time
import heapq

class Process:

    def __init__(self, pid, total):
        self.pid = pid
        self.total = total
        self.clock = 0
        self.queue = []
        self.reply_count = 0

    def request_cs(self):
        self.clock += 1
        timestamp = self.clock
        print(f"Process {self.pid} requesting CS at time {timestamp}")
        heapq.heappush(self.queue, (timestamp, self.pid))
        return timestamp

    def receive_request(self, timestamp, pid):
        self.clock = max(self.clock, timestamp) + 1
        print(f"Process {self.pid} received request from P{pid}")
        heapq.heappush(self.queue, (timestamp, pid))

    def send_reply(self, pid):
        print(f"Process {self.pid} sends REPLY to P{pid}")

    def receive_reply(self):
        self.reply_count += 1

    def enter_cs(self):
        if self.reply_count == self.total - 1:
            print(f"Process {self.pid} ENTERS critical section")
            time.sleep(1)
            print(f"Process {self.pid} EXITS critical section\n")
            heapq.heappop(self.queue)

processes = [Process(i,3) for i in range(3)]
timestamps = []

for p in processes:
    timestamps.append(p.request_cs())

for i in range(3):
    for j in range(3):
        if i != j:
            processes[j].receive_request(timestamps[i], i)
            processes[j].send_reply(i)

for p in processes:
    for _ in range(2):
        p.receive_reply()

for p in processes:
    p.enter_cs()
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                    zsh - expt_5

➜ expt_5 python3 lamport.py
Process 0 requesting CS at time 1
Process 1 requesting CS at time 1
Process 2 requesting CS at time 1
Process 1 received request from P0
Process 1 sends REPLY to P0
Process 2 received request from P0
Process 2 sends REPLY to P0
Process 0 received request from P1
Process 0 sends REPLY to P1
Process 2 received request from P1
Process 2 sends REPLY to P1
Process 0 received request from P2
Process 0 sends REPLY to P2
Process 1 received request from P2
Process 1 sends REPLY to P2
Process 0 ENTERS critical section
Process 0 EXITS critical section

Process 1 ENTERS critical section
Process 1 EXITS critical section

Process 2 ENTERS critical section
Process 2 EXITS critical section

➜ expt_5
```

Ricart–Agrawala Algorithm:

ricart.py Python

```python
import time

class Process:

    def __init__(self, pid, total):
        self.pid = pid
        self.total = total
        self.clock = 0
        self.reply_count = 0

    def request_cs(self):
        self.clock += 1
        print(f"Process {self.pid} requests CS at time {self.clock}")
        return self.clock

    def receive_request(self, timestamp, pid):
        self.clock = max(self.clock, timestamp) + 1
        print(f"Process {self.pid} received request from P{pid}")
        self.send_reply(pid)

    def send_reply(self, pid):
        print(f"Process {self.pid} sends REPLY to P{pid}")

    def receive_reply(self):
        self.reply_count += 1

    def enter_cs(self):
        if self.reply_count == self.total - 1:
            print(f"Process {self.pid} ENTERS critical section")
            time.sleep(1)
            print(f"Process {self.pid} EXITS critical section\n")



processes = [Process(i,3) for i in range(3)]

timestamps = []

for p in processes:
    timestamps.append(p.request_cs())

for i in range(3):
    for j in range(3):
        if i != j:
            processes[j].receive_request(timestamps[i], i)

for p in processes:
    for _ in range(2):
        p.receive_reply()

for p in processes:
    p.enter_cs()
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                    zsh - expt_5

➜ expt_5 python3 ricart.py
Process 0 requests CS at time 1
Process 1 requests CS at time 1
Process 2 requests CS at time 1
Process 1 received request from P0
Process 1 sends REPLY to P0
Process 2 received request from P0
Process 2 sends REPLY to P0
Process 0 received request from P1
Process 0 sends REPLY to P1
Process 2 received request from P1
Process 2 sends REPLY to P1
Process 0 received request from P2
Process 0 sends REPLY to P2
Process 1 received request from P2
Process 1 sends REPLY to P2
Process 0 ENTERS critical section
Process 0 EXITS critical section

Process 1 ENTERS critical section
Process 1 EXITS critical section

Process 2 ENTERS critical section
Process 2 EXITS critical section

➜ expt_5
```